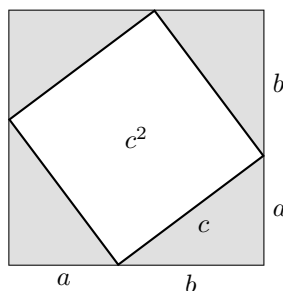


Proof of the Pythagorean Theorem

Proof

Consider a right triangle with legs of length a and b , and hypotenuse c .

We can construct a square with side length $a + b$ and place four copies of the right triangle inside it, each copy a quarter turn from the last, so that along every side of the large square a leg of length a is followed by a leg of length b :



The area of the large square is $(a + b)^2$.

The four right triangles each have an area of $\frac{1}{2}ab$.

The quadrilateral left in the center has all four sides equal to c , because each side is the hypotenuse of one copy. Its corners are right angles: where two copies meet on a side of the large square, that straight angle is made up of the two acute angles α and β of the triangle together with the corner of the quadrilateral, and $\alpha + \beta = 90^\circ$ since they are the acute angles of a right triangle. The center is therefore a square, and its area is c^2 .

Thus, we have:

$$(a + b)^2 = 4 \left(\frac{1}{2}ab \right) + c^2$$

Simplifying, we get:

$$a^2 + 2ab + b^2 = 2ab + c^2$$

Subtracting $2ab$ from both sides, we obtain:

$$a^2 + b^2 = c^2$$

This completes the proof of the Pythagorean theorem.